

Paper 29 - Monster/Universal Energy-Bound Hypotheses

CQE-paper-29

CQECMPLX Formal Paper Series

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Construction Basis

Record high-speculation finite-group/energy-bound ideas as candidate horizons with strict falsifier obligations.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-29.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-29\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-29\verify_monster_energy_bound_hypotheses.py
- receipt: production\formal-papers\CQE-paper-29\monster_energy_bound_hypotheses_receipt.json (CQE-paper-29: pass_with_quarantined_hypotheses)
- body: production\papers\CQE-paper-29\PAPER-BODY.md
- source: production\papers\CQE-paper-29\SOURCE.md
- formal: production\papers\CQE-paper-29\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-29\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-29\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-29.md

Paper 29 - Monster/Universal Energy-Bound Hypotheses

Abstract

Paper 29 is the corpus horizon paper for Monster and energy-bound language. It does not promote a universal energy bound. It closes only the proof rows that the package can actually replay: the arithmetic identity $196883 = 47 \cdot 59 \cdot 71$, the McKay decomposition $196884 = 1 + 196883$, and the finite VOA partition $Z(q) = 2q^0 + 6q^5$ on the eight chart states.

The paper's main scientific function is quarantine. Older recovered drafts contain stronger phrases such as Monster bound, Higgs maximum, and universal energy ceiling. In the production kernel these phrases are not discarded, but they are demoted to explicit hypothesis rows. Each row carries a falsifier and cannot become a theorem until a witness function supplies the missing transport from finite chart arithmetic to physical measurement.

Closed Evidence

The receipt is generated by ``production/formal-papers/CQE-paper-29/verify_monster_energy_bound_hypotheses.py``. It imports the live ``lattice_forge.centroid_voa`` surface and writes ``monster_energy_bound_hypotheses_receipt.json``.

The first closed row is integer arithmetic:

```
47 * 59 * 71 = 196883
196884 = 1 + 196883
```

The verifier checks the three factors as prime integers, checks the product, and checks the $1 + 196883$ decomposition. This is an arithmetic and reference row only. It is not a physical bound.

The second closed row is the finite chart partition. The verifier calls ``verify_voa_sector_decomposition`` and obtains:

```
Z(q) = 2q^0 + 6q^5
weight distribution = {0: 2, 5: 6}
```

The two weight-0 states are the true vacua $(0,0,0)$ and $(1,1,1)$. The other six states are the excited finite-chart orbit. The verifier also calls ``verify_centroid_voa_chain``, which preserves the same finite chart conclusion: two fixed points and six period-4 states. The chain conclusion explicitly states that any VOA or Moonshine identification requires an additional transport theorem.

Definitions

The active center ``C`` is the observer-selected chart center carried from Paper 00 through the suite. In this paper it is read through the finite ``lattice_forge.centroid_voa`` chart.

A VOA weight is the output of ``voa_weight(s)``, a wrap-step count assigned to a state ``s`` in the eight-state chart. It is an energy analog, not a calibrated energy.

The Monster representation dimension row is the integer 196883 . This paper uses it as a finite arithmetic reference and combinatorial ceiling analog. It does not assign physical units to it.

The McKay row is the integer decomposition $196884 = 1 + 196883$. This paper uses it as a reference decomposition, not as a new Moonshine theorem.

A horizon hypothesis is a useful candidate reading that is allowed to remain in the corpus only

if it is marked as unwitnessed and falsifiable.

A witness function is the missing function that would convert a horizon hypothesis into a closed theorem: it must name the input domain, output domain, units when relevant, verifier, receipt, and falsifier.

Claims

- The product $47 \times 59 \times 71$ equals 196883, and 196884 equals $1 + 196883$.
- The eight-state centroid VOA chart has finite partition $Z(q) = 2q^0 + 6q^5$.
- The centroid chain verifies two fixed points and six period-4 states as a finite chart identity.
- No closed row in this paper supplies physical units, physical energy, or a universal physical ceiling.
- Any reading that treats VOA weight as physical energy, the Monster dimension as an energy ceiling, or Pariah/Happy-Family closure as a physical bound remains a quarantined hypothesis.
- A later paper or product may cite Paper 29 only by preserving this distinction between closed finite rows and open horizon rows.

Theorem 29

A Monster/energy-bound statement is valid in the CQECMPLX kernel only when it separates closed finite evidence from horizon interpretation: arithmetic and finite VOA rows may pass as proof rows, while every physical energy-bound reading remains an explicitly falsifiable hypothesis until a witness function exists.

Proof

Run `verify_monster_energy_bound_hypotheses.py`.

The arithmetic checks pass because the verifier multiplies the three listed factors and obtains 196883, then adds the observer row 1 and obtains 196884. The result closes an integer receipt. No physical measurement is introduced in this check.

The VOA checks pass because `verify_voa_sector_decomposition` returns $Z(q) = 2q^0 + 6q^5$ and the distribution $\{0: 2, 5: 6\}$. The verifier then calls the centroid chain and confirms the same finite structure with two fixed points and six period-4 states.

The quarantine checks pass because the verifier enumerates three hypothesis rows. H1 rejects promotion of `voa_weight` to physical energy without a units-bearing transport. H2 rejects promotion of 196883 to a universal energy or state-count ceiling without a proved fingerprint-to-Monster map. H3 rejects promotion of Pariah/Happy-Family closure to a physical boundary law without an encoding-invariant boundary theorem. Each row has `witness_function_present = false` and `physical_units_present = false`.

Therefore the paper closes the arithmetic and finite chart rows while proving that the energy-bound readings remain open. This proves Theorem 29.

Worked Example

Begin with the eight chart states. Compute ``voa_weight(s)`` for each state. The two homogeneous states ``(0,0,0)`` and ``(1,1,1)`` have weight 0. The remaining six states have weight 5. The partition is therefore ``2q^0 + 6q^5``.

Next compute the Monster reference arithmetic:

```
47*59*71 = 196883
1 + 196883 = 196884
```

Now attempt the forbidden promotion: read the weight-5 sector as a physical energy gap or read ``196883`` as a universal ceiling. The verifier records no witness function, no units, and no physical calibration. The correct output is not a failed paper; it is a successful quarantine receipt with closed proof rows and open hypothesis rows.

Open Obligations

The physical-energy transport theorem is open. A future claim must supply a units-bearing map from the finite VOA weight to a measured physical quantity.

The Monster ceiling theorem is open. A future claim must supply a proved map from the corpus fingerprint to a Monster representation or state-space bound.

The Pariah/Happy-Family boundary theorem is open. A future claim must show that the boundary reading is encoding-invariant before it can be used as a physical law.

The recovered high-claim drafts remain useful as a horizon map, but their strongest energy-bound language is not production proof text.

Suite Role

Paper 29 protects the suite from overclaiming at the exact place where overclaiming is easiest. It lets later papers and products use Monster, Moonshine, Pariah, and energy language only through a receipt that says which rows are proven and which rows are still hypotheses.

Paper 30 inherits this discipline when it sweeps the papers into a ribbon. Paper 31 inherits it when the corpus reviews itself as an observer process. Paper 32 inherits it when the master suite is packaged for deployment.